25901056 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-BVOC (HONOURS) Regular EXAMINATION November 2025

First SEMESTER

Skill Development Components (SDC) - MG1SDCSEG100 - SCIENCE OF SOUND

(2025 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Very Short Questions

Answer any **10** questions

Each question carries **2** marks

1. State one factor that affects the refraction of sound. [U] / [CO1]
2. State the function of a VU meter in one line. [K] / [CO2]
3. Give a mixing action based on equal loudness curves (one line). [U] / [CO3]
4. State one condition required to perceive binaural beats. [K] / [CO4]
5. Name two auditory illusions. [K] / [CO4]
6. Given a short attack and long release in an ADSR;name an instrument whose envelope this may resemble. [U] / [CO1]
7. Describe a short procedure to use pink noise for room level matching. [U] / [CO2]
8. What is a critical band briefly? [K] / [CO3]
9. How does reverberation affect perceived distance of a sound? [U] / [CO4]
10. Define dynamic range. [K] / [CO2]
11. Explain briefly why sound is treated as a pressure wave in air. [U] / [CO1]

12. What SPL value is commonly cited as the threshold of pain? [K] / [CO2]

[2x10 = 20]

Part B

Short Questions

Answer any **6** questions

Each question carries **5** marks

13. Explain occupational exposure standards for sound (briefly cite typical dB/time limits) and their rationale. [U] / [CO3]

14. Explain ITD and ILD cues and how they jointly provide horizontal localization information. [U] / [CO4]

15. Explain how masking and noise floor affect useful dynamic range in recordings. [U] / [CO3]

16. Compare the harmonic content of sine, square, triangle and sawtooth waves, and explain why they sound different. [U] / [CO1]

17. Explain how alignment level and operating level differ and why both matter. [U] / [CO2]

18. Propose a brief instruction to teach society about preserving long-term hearing health in everyday life. [K] / [CO3]

19. Describe the Doppler effect and derive the qualitative pitch change when a source moves toward a listener. [U] / [CO4]

20. Describe the functions of the outer, middle and inner ear in the chain of acoustic transduction. [K] / [CO3]

21. Define partials and overtones and describe how they shape perceived tone color. [U] / [CO1]

[5x6 = 30]

Part C

Essay Questions

Answer any **2** questions

Each question carries **10** marks

22. Explain Diffusion and reverberation. Compare its effects on acoustics. [U] / [CO1]

23. Describe dBu, dBV, dBm, LUFS metering. [K] / [CO2]

24. Explain the working principle of Cochlea in depth. [K] / [CO3]

[10x2 = 20]